

Supplementary Material: Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Temporal Cohort Shift

Vishesh Panghal*

*^aDepartment of AI & DS, Poornima Institute of Engineering and Technology
(PIET), Jaipur, India*

Section S1: Comprehensive Cross-Institutional Factorial Matrix

Table S1 provides the complete $2 \times 2 \times 4$ cross-institutional factorial matrix on the official MIMIC-CXR-JPG test cohort ($N = 3,403$ frontal radiographs from 3,041 diagnostic studies across $N = 289$ unique patients). Both unweighted and positive-weighted binary cross-entropy (BCE) models are evaluated across DenseNet-121 and ResNet-50 architectures for all four thoracic abnormalities. For positive-weighted models, paired zero-parameter analytic logit recalibration ($z \mapsto z - \log w$) is evaluated alongside patient-clustered 1,000-replicate bootstrap 95% confidence intervals.

Section S2: Diagnostic Study-Level Sensitivity Analysis

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, Table S2 summarizes external validation performance aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ unique patients). In accordance with standard radiological diagnostic practice, multiple views acquired during a single examination are aggregated using maximum probability pooling ($\max_i p_i$, where detection on any view indicates study-level positive status). Supplementary Table S2 evaluates

*Corresponding author.

Email address: vishesh@poornima.org (Vishesh Panghal)

Table S1: Supplementary Table S1: Complete 16-Row Cross-Institutional External Validation Matrix on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Radiographs, 3,041 Studies, 289 Unique Patients). Evaluated on 3-Seed Probability Ensembles with Image Prevalence (π), Empirical Null Reference Baselines ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), Patient-Clustered Bootstrap 95% Confidence Intervals, Brier Skill Scores ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and Paired Analytic Recalibration ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$).

Architecture	Objective	Pathology	Prevalence (π)	$\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.688 [0.656, 0.717]	0.2488	–	-0.140
DenseNet-121	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.523 [0.491, 0.556]	0.2375	–	-0.224
DenseNet-121	Unweighted BCE	Pneumonia	10.05%	0.0904	0.607 [0.560, 0.650]	0.0986	–	-0.091
DenseNet-121	Unweighted BCE	Edema	21.33%	0.1678	0.562 [0.524, 0.596]	0.2124	–	-0.266
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Pos-Weighted BCE	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.677 [0.645, 0.709]	0.2430	–	-0.113
ResNet-50	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.519 [0.489, 0.548]	0.2333	–	-0.203
ResNet-50	Unweighted BCE	Pneumonia	10.05%	0.0904	0.583 [0.540, 0.623]	0.0982	–	-0.086
ResNet-50	Unweighted BCE	Edema	21.33%	0.1678	0.516 [0.480, 0.552]	0.2126	–	-0.266
ResNet-50	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Pos-Weighted BCE	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

all 16 architecture-loss-pathology combinations with patient-clustered 1,000-replicate bootstrap 95% confidence intervals. As shown, study-level results demonstrate that all primary findings are invariant to whether the statistical unit is the individual radiograph or the diagnostic study: unweighted models maintain robust discrimination across views, while analytic logit recalibration ($z - \log w$) on positive-weighted models yields statistically significant reductions in study-level Brier error for Cardiomegaly ($\Delta\text{Brier} = -0.0601$ on DenseNet-121 and -0.0642 on ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0100$ and -0.0163 , both intervals strictly below zero), alongside consistent point-estimate reductions for Pleural Effusion (-0.0102 and -0.0118).

Section S3: Reproducibility and Code Execution Guide

The complete codebase is organized to support deterministic replication from raw prediction archives through all tables, figures, and statistical tests:

1. **Environment Setup:** Python 3.10+ with PyTorch, torchvision, scikit-learn, scipy, pandas, numpy, and matplotlib.
2. **Stage 3 Recalibration Execution:**

python3 scripts/run_stage3_recalibration.py

Computes probability ensembles ($\bar{p}_m = \frac{1}{3} \sum_{s=1}^3 p_{m,s}$) across all 8,302 prospective radiographs (16,604 paired model evaluations) for raw, temperature-scaled, analytic offset ($z - \log w$), fitted offset ($z + a^*$), and

Table S2: Supplementary Table S2: Diagnostic Study-Level Sensitivity Analysis on the Official MIMIC-CXR-JPG Test Cohort ($N = 3,041$ Studies across $N = 289$ Unique Patients). Multi-image studies are aggregated using maximum probability pooling ($\max_i p_i$). Evaluated on 3-seed probability ensembles with patient-clustered 1,000-replicate bootstrap 95% confidence intervals, local null Brier reference ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), and paired recalibration contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$).

Architecture	Objective	Pathology	Study Prevalence	Study $\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.695 [0.663, 0.723]	0.2488	–	-0.133
DenseNet-121	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.526 [0.497, 0.559]	0.2378	–	-0.219
DenseNet-121	Unweighted BCE	Pneumonia	10.16%	0.0913	0.608 [0.562, 0.650]	0.0996	–	-0.091
DenseNet-121	Unweighted BCE	Edema	21.64%	0.1697	0.565 [0.528, 0.600]	0.2157	–	-0.271
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	32.55%	0.2196	0.674 [0.643, 0.704]	0.2556 $\rightarrow$ 0.2453	-0.0102 [-0.0427, +0.0236]	-0.164 $\rightarrow$ -0.117
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	26.54%	0.1951	0.527 [0.497, 0.558]	0.2933 $\rightarrow$ 0.2332	-0.0601 [-0.0814, -0.0367]	-0.503 $\rightarrow$ -0.195
DenseNet-121	Pos-Weighted BCE	Pneumonia	10.16%	0.0913	0.599 [0.555, 0.639]	0.1091 $\rightarrow$ 0.0992	-0.0100 [-0.0179, -0.0018]	-0.195 $\rightarrow$ -0.086
DenseNet-121	Pos-Weighted BCE	Edema	21.64%	0.1697	0.583 [0.550, 0.616]	0.1937 $\rightarrow$ 0.2165	+0.0228 [+0.0126, +0.0325]	-0.141 $\rightarrow$ -0.275
ResNet-50	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.682 [0.650, 0.714]	0.2431	–	-0.107
ResNet-50	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.521 [0.494, 0.548]	0.2340	–	-0.200
ResNet-50	Unweighted BCE	Pneumonia	10.16%	0.0913	0.584 [0.543, 0.624]	0.0992	–	-0.087
ResNet-50	Unweighted BCE	Edema	21.64%	0.1697	0.516 [0.480, 0.553]	0.2158	–	-0.272
ResNet-50	Pos-Weighted BCE	Pleural Effusion	32.55%	0.2196	0.625 [0.594, 0.655]	0.2685 $\rightarrow$ 0.2567	-0.0118 [-0.0427, +0.0251]	-0.223 $\rightarrow$ -0.169
ResNet-50	Pos-Weighted BCE	Cardiomegaly	26.54%	0.1951	0.523 [0.493, 0.552]	0.2858 $\rightarrow$ 0.2217	-0.0642 [-0.0852, -0.0434]	-0.465 $\rightarrow$ -0.136
ResNet-50	Pos-Weighted BCE	Pneumonia	10.16%	0.0913	0.588 [0.542, 0.627]	0.1155 $\rightarrow$ 0.0992	-0.0163 [-0.0267, -0.0068]	-0.265 $\rightarrow$ -0.086
ResNet-50	Pos-Weighted BCE	Edema	21.64%	0.1697	0.571 [0.538, 0.606]	0.1951 $\rightarrow$ 0.2164	+0.0213 [+0.0100, +0.0330]	-0.149 $\rightarrow$ -0.275

Platt-scaled models, running 2,000-replicate patient-clustered bootstrap inference.

3. Publication Artifact Compilation:

`python3 scripts/compile_publication_artifacts.py`

Generates Tables 1–4 in LaTeX and markdown formats, alongside publication figures (Figures 2–6).

4. Verification Audit:

`python3 scripts/audit_cibm.py`

Verifies all formatting, word count, highlight length (≤ 85 characters), and statutory compliance rules for Computers in Biology and Medicine.